

SLIP – 2

Section I: Data Structures [15 Marks]

Q.1) Write a C program to implement static Implementation of Queue of integers with following operation:

Answer:

```
#include <stdio.h>
#define MAX 100

int q[MAX], front, rear;

void initialize()
{
    front = 0;
    rear = -1;
}

int isEmpty()
{
    return front > rear;
}

int isFull()
{
    return rear == MAX - 1;
}

void insert(int x)
{
    if(isFull())
        printf("Queue Overflow\n");
    else
        q[++rear] = x;
}

void delete()
{
    if(isEmpty())
        printf("Queue Underflow\n");
    else
        printf("Deleted element = %d\n", q[front++]);
}

void display()
{
    int i;
```

```

        if(isEmpty())
printf("Queue is empty\n");    else
{
        for(i = front; i <= rear; i++)
printf("%d ", q[i]);          printf("\n");
    }
}

int main()
{
    int ch, x;

    initialize();

    do
    {
        printf("\n1. Insert\n");
printf("2. Delete\n");          printf("3.
Check Empty\n");              printf("4.
Display\n");                  printf("5. Exit\n");
printf("Enter your choice: ");
scanf("%d", &ch);

        switch(ch)
        {
            case 1:
                printf("Enter element: ");
scanf("%d", &x);                insert(x);
break;                    case 2:
delete();                  break;
            case 3:
                if(isEmpty())
printf("Queue is empty\n");          else
printf("Queue is not empty\n");
break;                    case 4:
display();
break;                    case 5:
break;
            default:
                printf("Invalid choice\n");
        }
    } while(ch != 5);

    return 0;
}

```

Section II: Database Management Systems [15 Marks]

Q 2) Consider the following database(primary keys are underlined).

Employee(en^o,en^a,des^{ignation},sal^{ary})

Department(dno,dname,location)

The relationship is as follows: Employee-Department: many-to-one.

Q 2(A)

A. Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```
CREATE TABLE Department(  
  dno INT PRIMARY KEY,      dname  
  VARCHAR(50),      location  
  VARCHAR(50)  
);  
  
CREATE TABLE Employee(  
  eno  
  INT PRIMARY KEY,      ename  
  VARCHAR(50),      designation  
  VARCHAR(50),      salary  
  NUMERIC(10,2),  
  dno INT REFERENCES Department(dno)  
);  
  
INSERT INTO Department VALUES  
(1, 'HR', 'Nashik'),  
(2, 'Sales', 'Delhi'), (3, 'IT', 'Pune');  
  
INSERT INTO Employee VALUES  
(101, 'Sneha', 'Manager', 50000, 1),  
(102, 'Rahul', 'Clerk', 20000, 2),  
(103, 'Meena', 'Developer', 40000, 3);
```

Q 2(B)

B. Execute the following queries in PostGreSQL (any 3) [6 marks]

i. Give a 5% raise in salary to all the employees.

Answer:

```
UPDATE Employee  
SET salary = salary * 1.05;
```

ii. Display average salary of an employee.

Answer:

```
SELECT AVG(salary) FROM Employee;
```

iii. List the details of all the departments located at 'Nashik' city.

Answer:

```
SELECT * FROM Department  
WHERE location = 'Nashik';
```

iv. Display the details of employees whose names ends with an alphabet 'a'.

Q 2(C)

C. Write a query to list the details of employees who do not work in any of the departments located at 'Delhi'. [4 marks]

Answer:

```
SELECT * FROM Employee E  
WHERE NOT EXISTS  
(  
    SELECT 1 FROM Department D  
    WHERE D.dno = E.dno  
    AND D.location = 'Delhi'  
);
```

Q 3) Viva [5 marks]